

ABDELJAWAD MOHAMMED

Software Engineer | Automation & Systems Integration

Abdeljawad202125001@gmail.com • +20 114 249 4264 • Cairo, Egypt

[Portfolio: abdeljawad-mohammed.github.io/portfolio](https://abdeljawad-mohammed.github.io/portfolio) • [LinkedIn: linkedin.com/in/abdeljawad-mohammad](https://www.linkedin.com/in/abdeljawad-mohammad)

PROFILE

Software Engineer with 2+ years of experience designing, building, and operating production systems across e-commerce, ERP, infrastructure, and business operations. Experienced in backend development, system integration, automation, internal tools, AI-powered applications, and production infrastructure from requirements through deployment and ongoing support. Strong hands-on background across Python, JavaScript, APIs, cloud infrastructure, DevOps, and e-commerce platforms, with a focus on reducing manual work, improving operational efficiency, and building reliable systems around real business workflows.

PROFESSIONAL EXPERIENCE

Software Engineer — Automation & Integrations, Locafy 03/2025 – Present | Cairo, Egypt

- Self-hosted and operated the company automation infrastructure across AWS EC2 and Namecheap VPS, running 9 production workloads with Docker, nginx, systemd, and Let's Encrypt; led a verified cross-provider migration with parallel rebuild and no downtime.
- Engineered Magento → Odoo order automation on self-hosted n8n, reducing order-to-quotation time from up to 24 hours to ~2 minutes and eliminating manual order entry.
- Led the Magento → Shopify replatform of the production store, migrating 1,122 products through the Shopify GraphQL Admin API with configurable-to-variant mapping, price/inventory handling, and Arabic storefront support.
- Built Shopify → Odoo integration with a Rasayel WhatsApp confirmation gateway: Shopify webhooks create Odoo draft quotations, while customer yes/no responses automatically confirm or cancel orders across both systems.
- Architected the Locafy Intelligence Platform, a Node.js + Express service integrating 9 business systems including Odoo, Meta, GA4, Search Console, Clarity, PageSpeed, Sentry, Rasayel, and Anthropic Claude to generate automated CEO reports and deliver PDFs via WhatsApp.
- Built Locafy Shipments Operations, a 1,950-line logistics tool integrating Magento and Bosta Courier API v2, reducing shipment processing from ~5 minutes to ~20 seconds per order with bilingual address matching and automated Excel reporting.
- Built Node.js automation against the Meta Marketing API for 155 campaigns, 269 ad sets, and 365 ads, covering campaign discovery, account-state validation, reporting, and programmatic draft creation.
- Developed and maintained Odoo integrations and customizations across CRM, Inventory, and Sales using XML-RPC, JSON-RPC, Odoo Studio, and Python automation.

Software Engineer — E-commerce & Business Systems, Brass & Wood 06/2024 – Present | Cairo, Egypt

- Designed and productionized a self-hosted visual search engine over 30,158 Odoo and Magento product vectors using CLIP ViT-B/32, fastembed/ONNX Runtime, Qdrant, and rembg, with multi-angle indexing and product-level deduplication.
- Engineered Odoo → Magento product creation with custom Odoo Studio models and fields plus a Python worker that creates configurable products and variants, synchronizes processing status, and handles Magento product-state edge cases.
- Architected a company-wide automated backup system protecting 662 GB across 168,333 files using scheduled VPS exports, a 5 TB Hetzner off-site vault with daily snapshots, an offline self-waking local backup node, and Telegram monitoring.
- Built BW Telegram Vendor Groups, an Odoo-to-Telegram manufacturing coordination system that creates groups per sales order × workshop, distributes item documents and photos, tracks Arabic production stages and delays, and triggers SLA alerts.
- Built BW Operations Help, an Arabic warehouse application for SKU/barcode/name lookup with live Odoo data covering product images, stock across 9 warehouses, sales orders, purchase orders, vendors, and inventory movements.

- Built and maintained the BW Price & Status Automation Suite, using scheduled Python services and systemd timers to synchronize pricing and product status between Odoo and Magento every 10–60 minutes.
- Implemented Magento multi-store expansion across 10 countries and 20 store views, including a GeoNames → MySQL pipeline for country, region, and city data plus store-specific routing and localization.
- Automated product-image processing with background removal, AI background replacement, and brass color-variant generation, integrating n8n, Google Drive, and AI image APIs to reduce manual image-editing work from hours to seconds.
- Prototyped a browser-based 3D room planner using Three.js, allowing customers to place real Brass & Wood products into custom room layouts with configurable walls, floors, lighting, doors, and windows, backed by an AI image-to-3D model pipeline.

TECHNICAL SKILLS & TOOLS

Automation & Integration

n8n · Webhooks · REST · GraphQL · XML-RPC · JSON-RPC · Magento REST · Odoo XML-RPC · Shopify Admin API · Rasayel GraphQL · Meta Marketing API · Telegram Bot API · Telethon / MTProto · Bosta Courier API · ClickUp API

Cloud & DevOps

AWS EC2 · RDS · S3 · IAM · Route 53 · VPS / Linux Administration · Docker · nginx · Apache · systemd · cron · Let's Encrypt / certbot · SSH · rclone · mysqldump · Cloudflare Pages / Tunnels · Netlify · Vercel · Railway / Render

Languages

Python · JavaScript (ES2022 / ES Modules) · TypeScript · PHP · SQL · HTML5 · CSS3 / LESS / SCSS · Bash · PowerShell · Liquid · C · Java

Frontend & Visualization

Vanilla JavaScript · React · Tailwind CSS · Three.js · Chart.js · Recharts · GSAP · ExcelJS · jsPDF / html2pdf · Framer Motion · Zustand

Development & Monitoring Tools

Git · VS Code · Claude Code · Playwright · Puppeteer · npm · pip / venv · Postman · Maven · GCC · draw.io · Google Analytics 4 · Search Console · PageSpeed Insights · Sentry · Microsoft Clarity · Hotjar

Backend & APIs

Node.js · FastAPI · Express · Next.js · uvicorn · REST APIs · GraphQL · XML-RPC · JSON-RPC · Webhooks · OAuth2

AI & Machine Learning

Anthropic Claude API · Multi-Agent Workflows · Structured Outputs · CLIP ViT-B/32 · fastembed · ONNX Runtime · Qdrant Vector Search · rembg · fal.ai · Rodin · Meshy · Tripo3D · Tesseract OCR · Blender

Databases & Data

MySQL · PostgreSQL · SQLite · Supabase · Qdrant · Prisma · Google Sheets · JSON / JSONL / CSV

Platforms

Magento 2 · Odoo · Shopify · n8n · Rasayel · Meta Marketing API · Telegram · Google Workspace · ClickUp · Workiom

Professional Strengths

Technical Ownership · Problem Solving · Cross-Functional Collaboration · Stakeholder Communication · Production Troubleshooting · Documentation

TECHNICAL PROJECTS & CONTRIBUTIONS

Self-Hosted Automation Infrastructure & Cross-Provider Migration

Migrated 9 production workloads from AWS EC2 to a Namecheap VPS using verified tarball backups and a parallel rebuild, preserving Docker services, nginx routing, systemd workloads, HTTPS, firewall rules, and application state without downtime. (AWS EC2 · Namecheap VPS · Docker · nginx · systemd · Bash · UFW · Let's Encrypt)

Company Backup & Disaster Recovery System

Architected and deployed a company-wide automated backup system across Odoo, Google Drive, Rasayel, code, and infrastructure, protecting 662 GB across 168,333 files through scheduled VPS exports, a 5 TB Hetzner off-site vault with daily snapshots, and an offline self-waking local backup node with Telegram status reporting.

Brassandwood Visual Search Engine

Built a self-hosted visual product search engine indexing 30,158 Odoo and Magento product vectors, enabling image and semantic similarity search using CLIP embeddings, Qdrant nearest-neighbor retrieval, multi-angle indexing, and background-removal preprocessing. (Python · FastAPI · CLIP ViT-B/32 · fastembed · ONNX Runtime · Qdrant · rembg)

Magento → Shopify Store Migration

Built the migration pipeline for a production Magento-to-Shopify replatform, importing 1,122 products with configurable-to-variant mapping, price and inventory handling, and supporting English/Arabic storefront data through Shopify APIs. (Python · Shopify GraphQL Admin API · Shopify Translations API · Magento REST · OAuth)

BW Sales Dashboard

Built an 8-section sales intelligence application combining Odoo, Meta Ads, and Rasayel data with drill-down analytics, targets and forecasts, multi-currency conversion, customer and product analysis, Excel export, and protected commissions reporting. (Python · Vanilla JavaScript · Chart.js · Odoo XML-RPC · Meta Marketing API · Rasayel GraphQL)

Locafy Shipments Operations

Built shipment automation tool reducing per-order processing from 5 minutes to 20 seconds — fuzzy Arabic/English district matching, 2-click shipment creation via Bosta API, real-time Google Sheets sync. (Vanilla JS · n8n · Bosta API)

Manufacturing Cost Estimator

Built a manufacturing cost-estimation application using live Odoo product and supplier data with a comparables-based pricing model, vendor-price analysis, dataset generation, and hold-out validation. (Python · FastAPI · Odoo XML-RPC · Data Analysis)

Magento → Odoo Order Sync

Engineered webhook-based automation syncing orders from Magento to Odoo on checkout — 100% manual entry eliminated, Hours reduction in fulfillment lead time. (Magento API · Odoo · n8n)

Shopify → Odoo + WhatsApp Confirmation Gateway

Built an end-to-end order confirmation workflow where Shopify webhooks create Odoo draft quotations and Rasayel WhatsApp templates collect customer approval or rejection, automatically confirming or cancelling the order across both systems. (n8n · Shopify Webhooks · Odoo JSON-RPC · Rasayel GraphQL · WhatsApp)

WhatsApp Sales Assignment Bot

Built lead routing bot evaluating region and interest for auto-assignment — reduced response time from 45 min to 2 min, sales efficiency up 30%. (Rasayel API · n8n)

Product Image Automation

Built AI-powered system for background removal and color variant generation — reduced hours of manual editing to seconds per SKU. (n8n · AI Image API)

Locafy Insights Dashboards

Built real-time Magento and Shopify analytics dashboards covering 1,000+ live orders with revenue, brand, city, category, cancellations, returns, and status-transition analysis for daily operations monitoring. (Vanilla JavaScript · Magento REST · Shopify Admin API · n8n)

BW Telegram Vendor Groups

Built an Odoo-to-Telegram manufacturing coordination system that automatically creates one group per sales order and workshop, distributes product documents and images, tracks Arabic production-stage updates, records delays, triggers SLA alerts, and feeds a secured management dashboard. (FastAPI · Telethon · Telegram Bot API · Odoo XML-RPC · Google Drive · systemd)

Google Workspace Migration

Led full company data migration to Google Workspace Shared Drives using MultCloud.

Odoo → Magento Product Creation Pipeline

Engineered an Odoo-to-Magento product publishing workflow with custom Odoo Studio models and fields, a push-to-Magento interface, and a Python worker that creates configurable products and variants while synchronizing processing status back to Odoo. (Python · Odoo Studio · Odoo XML-RPC · Magento REST)

Meta Ads Automation

Built Node.js automation around the Meta Marketing API for an account containing 155 campaigns, 269 ad sets, and 365 ads — covering campaign structure discovery, account-state validation, reporting, and programmatic draft-campaign creation. (Node.js · Meta Marketing API · REST · System User Authentication)

BW Operations Help — Warehouse Lookup

Built an Arabic warehouse operations app allowing staff to search by SKU, barcode, or Arabic product name and instantly retrieve product images, stock across 9 warehouses, sales orders, purchase orders, vendors, and inventory movements from Odoo. (Python · Odoo XML-RPC · HTML · CSS · systemd · nginx)

3D Room Planner & AI 3D Model Pipeline

Prototyped a browser-based 3D interior planner supporting custom room geometry, draggable real products, doors, windows, lighting, floors, walls, and project export, backed by an AI image-to-3D pipeline evaluated across Rodin, Meshy, and Tripo3D with Blender rendering. (Three.js · GLTFLoader · fal.ai Rodin · Meshy · Tripo3D · Blender · JavaScript)

BW Price & Status Automation Suite

Built and deployed scheduled synchronization services keeping Odoo and Magento product pricing and status data aligned through three independent systemd timers running at 10–60 minute intervals. (Python · Odoo XML-RPC · Magento REST · systemd · Linux)

Brassandwood B2B Trade Portfolio

Built a 13-page B2B trade catalog from live Magento product data with bilingual-ready structured content, 47 real products across 8 collections, automated catalog generation, and Cloudflare Pages deployment. (HTML · CSS · Vanilla JavaScript · Magento REST · Node.js · Cloudflare Pages)

Automated Product Specification PDF Generator

Built a one-click system for generating branded product specification sheets from structured product information, enabling reusable customer-facing PDFs without manual graphic-design work. (Python · HTML · CSS · html2pdf.js · JavaScript)

Locafy Intelligence Platform

Built a Node.js + Express intelligence platform integrating 9 business systems including Odoo, Meta, GA4, Search Console, Clarity, PageSpeed, Sentry, Rasayel, and Anthropic Claude — generating AI-powered CEO briefings, branded PDFs, and automated WhatsApp delivery. (Node.js · Express · Anthropic SDK · Puppeteer · node-cron · REST APIs)

EDUCATION

MSc in Computer Science , <i>Cairo University - CU</i>	10/2024 – Present Cairo, Egypt
BSc in Computer Science , <i>October 6 University - O6U</i>	09/2020 – 06/2024 October, Egypt
- CGPA 3.71 / 4.0 · Graduated with Excellent Grade. - Graduation Project: AI Ducted AC Control System (Python, Fuzzy Logic) · Grade: Excellent.	

LANGUAGES

Arabic	English
Native	Professional Working Proficiency